

Aliah University

Odd Semester Examination(Autumn Semester) 2022

(For 2nd Year 3rd Semester B.Tech CSE)

Paper Name: Data Structures & Algorithms

Paper Code: CSEUGPC01

Full Marks:80

Time: 3 hours

Group A

Answer all the questions

(5X2=10)

1. Define Dequeue.
2. Differentiate linear and nonlinear data structure with examples.
3. What is sparse matrix?
4. What is threaded binary tree?
5. Define full binary tree and complete binary tree.

Group B

Answer any 5 questions

(5X6=30)

1. Give algorithms to perform following operations in a singly linked list.(3+3)
 - (a) Insert a new node after a given node.
 - (b) Delete last node.
2. Differentiate queue and circular queue. How is queue represented as a linked list?(3+3)
3. Compare and contrast singly linked list and doubly linked list. How will you represent header node in a Linked List?(3+3)
4. What do you understand by complexity of an algorithm? Write worst case and best case complexity of linear search.(3+3)
5. Let LIST be a singly linked list in memory. Write an algorithm to find number of times a given data item called ITEM occurs in LIST.(6)
6. Differentiate between an array and a stack. Consider a two dimensional array A of order [25X4].The base address of the array is 400,words per memory cell is 4.Find the address of A[12,4] using row major and column major addressing.(3+3)

Group C

Answer any 4 questions

(4X10=40)

1. How stack can be represented using a linked list? Give the algorithms for PUSH and POP operations using stack as a linked list. Evaluate the following postfix expression $ABC*D/+$ where $A=2$ $B=3$ $C=4$ $D=6$.
(2+6+2)

2. Define binary search tree. Write an algorithm to search an element from a binary search tree. Suppose that the following list of letters is inserted in order into an empty binary search tree:

J, R, D, G, T, E, M, H, P, A, F, Q

i) Find the final tree T.

ii) Find the pre order, in order and post order traversal of T.

iii) What is the height of the created binary tree?

(2+3+5)

3. Give an algorithm to perform binary search. Using the algorithm, search for elements 23 and 47 in the given set of elements [12 23 27 35 39 42 50]. Compare binary search and linear search. (3+5+2)

4. Write algorithm for Insertion sort. Discuss the complexity of Insertion sort. Illustrate the insertion sort algorithm on input [30, 20, 10, 60, 70, 40]
(4+3+3)

5. Define collision. What is linear probing? What are the properties of a good hash function? The following keys 10, 16, 11, 1, 3, 4, 23 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table?
(2+2+2+4)